

Revised 2026-09-10 | Deliverable No. 10 | English + 简体中文

# Backend Operating Manual - ellis-visa.com

## Topology

Hetzner CX23 (Falkenstein). Caddy terminates TLS for ellis-visa.com and serves the static app from /opt/ellis/src/renderer/dist; /api/\* is proxied to the FastAPI backend (systemd unit ellis-backend) on 127.0.0.1:8000. Data: SQLite at /var/lib/ellis/ellis.db.

## Routine operations

- Status: `systemctl status ellis-backend caddy`
- Logs: `journalctl -u ellis-backend -f (backend), -u caddy (edge)`
- Deploy a tested release pinned to its commit and artifact checksum. Verify the current database backup and retain the previous release before updating. Record the release identifier, migration steps and rollback procedure; smoke test the backend, public application and quality console after restarting.
- Health: `GET https://ellis-visa.com/api/health`

## Data protection

Daily full database backup is required. `backend/scripts/backup_database.py` uses SQLite's online backup mechanism to include committed WAL content, validates the destination database and atomically publishes a private (mode 0600) file only after a successful integrity check. The daily wrapper retains the existing `/var/backups/ellis/ellis-<weekday>.db` paths. Never fall back to copying a live database file when SQLite backup fails. Record successful backups and failures; verify the deployed daily wrapper and its actual output, rather than inferring success from a schedule's existence. Keep a verified off-host copy for host-loss recovery. This is a database backup; release artifacts, configuration and necessary external storage require their own recovery arrangements.

For restore, stop the service, preserve the current state, restore the verified backup using the release-specific runbook, and check database integrity and public/quality behavior before reopening service. The contractual recovery target is at most one hour; a dated timed full recovery exercise is required to prove it. `systemd` process restart is useful but is not a disaster-recovery test. The 2026-09-10 live database-only drill restored a 23,834,624-byte backup, 99 tables and 949 route-table rows with integrity ok; database backup and restore took 0.708 seconds. The daily wrapper was updated. Full-service recovery, including host/app/configuration/secrets/DNS and off-host durability, is not certified by that database check. Details are in `AVAILABILITY.md`.

## Quality-control backend

<https://ellis-visa.com/#ops> - records with combined spot-check filters (names, codes and Chinese accepted), per-record 25-field checklists, confidence and source-substantiation chips, one-click error flagging into the tracked correction queue, the change log, freshness, and the two-sheet Excel export.

The current confidence display is binary High/Low, with AI/human authorship kept in source provenance and publication controlled separately by the existing evidence/conflict gates. See

`DATA_CALIBER_MANUAL.md` for precise definitions. `info_validity` is the published policy end date only; unknown dates remain blank. `freshness_valid_until` is separate internal review-deadline metadata.

The user-approved completeness target includes documented Not applicable/Not published across all 25 fields; unknown/unsupported/pending states stay incomplete. Use the latest dated release audit and its immutable snapshot SHA for current counts. Recompute documented completeness across all 25 contractual fields; retain operational applicable 20 and literal 25 non-null diagnostics separately. A previous release's percentage must not be presented as current. Completion, source-link presence and publication availability do not certify policy accuracy.

The current bilingual HTML/PDF manuals have been regenerated for these semantics. Old PASS labels, sample availability percentages, or unmeasured recovery claims are not current acceptance evidence.

## Security posture

UFW allows 22/80/443 only. Backend binds loopback; only Caddy is exposed. Secrets ship encrypted (backend/secrets.enc, AES-256-CBC) and are decrypted on the machine at deploy time; backend/.env never enters git.

## Route and product publication

QC keeps every product available for inspection. Product held and route\_held are distinct: a partial route can publish a supported exemption or required e-visa while separate optional products remain withheld. The public page labels this state and excludes unsupported alternative claims from its answer and chat. Fees, filing deadlines and required conditions cannot be discarded to make a route publishable. Report default availability, fully released routes and held products separately; partial publication is not 100% product coverage.

## Source corrections and issue dispositions

Reviewed releases preserve raw source history, issue records and prior review dates. Changed fields receive their actual scoped verification date; unrelated dates are not renewed. Before installing a source correction, compare the captured raw cache, merged answer, route row, provenance, ordered seed overlays and operator overrides. Rebuild the exact approved candidate from those six layers. Pause writers and recheck those preconditions at installation; changed input requires a new review. A narrowly scoped correction must not renew unrelated fields or release products whose independent evidence or disputes are unresolved.

GET /api/database/issues returns revision\_sha256 for each current issue. Clients that bind their review to that revision pass expected\_issue\_sha256 with the disposition request. The API also checks its own

read-to-write revision for all callers. A changed issue returns HTTP 409. The issue update and AuditEvent commit in one transaction. Older clients that omit the expected revision do not receive protection for changes between their earlier screen review and the API request. Separate route/provenance prechecks are not an atomic route-row lock. After an uncertain POST result, inspect current state before deciding what to do; do not blindly resubmit. Retain the actual actor, review note and source evidence.

## Review scheduling priorities

After eligibility and continuation filters, review the four reported critical routes first, then US/Hong Kong routes to the 18 station markets plus mainland China, then other station routes to those markets, then the remaining inventory. These are user and project priorities, not measured traffic statistics. Every eighth dispatch takes the eligible route with the oldest recorded source-check attempt; missing or invalid attempt dates sort first. This prevents indefinite starvation without using policy expiry as the priority clock. Continuation retains the dispatch position and original cycle deadline. A release must not reset the five-hour budget or count unattempted work as verified.

## Acceptance evidence

Keep the supplied requirements ledger with its original document hashes and clause identities. A release report must distinguish implemented behavior, measured checks and remaining external requirements. A scheduled report is not proof that the required human/provider review occurred; a short database restore does not establish full service recovery, and a short monitoring interval cannot prove a monthly 99.99% availability target. Retain actual dated evidence and record any remaining acceptance item without inventing sign-off.

Source: BACKEND\_OPERATING\_MANUAL.md | SHA-256  
c1bf3c76a1ced68587677ee9526dd78763d21cb7b60d05494354f9f0771d3f4c

# Ellis 信息质量管控后台操作手册

本手册记录当前实现及操作要求，不代表全部数据已经核实或项目验收已经通过。

## 系统结构

生产架构使用 Hetzner CX23 (Falkenstein)，由 Caddy 为 ellis-visa.com 终止 TLS，并提供 /opt/ellis/src/renderer/dist 中的静态应用。/api/\* 请求转发到本机 127.0.0.1:8000 的 FastAPI 服务 (systemd 服务名 ellis-backend)。SQLite 数据库路径为 /var/lib/ellis/ellis.db。

## 日常运行与发布

- 状态检查：systemctl status ellis-backend caddy。
- 后端日志：journalctl -u ellis-backend -f；边缘代理日志：journalctl -u caddy。
- 发布应使用经过测试、按提交与产物校验和固定的版本。更新前核实当前数据库备份，保留上一版本，并记录版本号、迁移步骤和回滚流程。重启后检查后台接口、公开应用与质控台。
- 健康检查地址：https://ellis-visa.com/api/health。

## 数据保护与恢复

每日必须进行完整数据库备份。backend/scripts/backup\_database.py 使用 SQLite 在线备份接口，包含已提交的 WAL 内容；对备份执行完整性校验后，才以原子方式发布权限为 0600 的私有文件。每日执行脚本沿用 /var/backups/ellis/ellis-<weekday>.db 路径。

SQLite 备份失败时，不得退回到复制运行中的数据库文件。应记录备份成功与失败，核查线上每日任务及实际产物；任务存在不代表备份成功。应保存已验证的异机副本，以支持主机丢失后的恢复。此机制只备份数据库；发布产物、配置与必要的外部存储须分别安排恢复。

恢复时先停止服务并保留当前状态，按该版本的恢复流程还原已验证的备份，再检查数据库完整性、公开页面及质控功能，然后恢复服务。合同恢复目标为不超过 1 小时，是否达标须由带日期和计时记录的完整恢复演练证明。systemd 自动重启进程有帮助，但不是灾难恢复验收证明。2026-09-10 实际数据库备份与还原用时 0.708 秒，23,834,624 字节、99 张表、949 条线路表记录，完整性正常；每日任务已更新。该演练不包含主机、应用、配置、密钥、DNS 或异机副本可靠性，不能证明完整服务恢复达标。

## 质控台与数据解读

浏览器打开 <https://ellis-visa.com/#ops>。记录页面提供可组合的抽查筛选（支持名称、代码与中文）、逐条 25 字段清单、置信度与信源支持标识、问题标记、可追踪纠错队列、变更记录、时效信息及双工作表 Excel 导出。

质控保留全部产品供检查。产品 held 与线路 route\_held 分别计数；部分发布的线路可以提供有证据的默认免签或需预先申请的电子签答案，同时拦截独立且未核实的可选签证。对客页面明确提示，并排除未核实的替代产品主张；不得为了发布而丢失费用、申请期限或入境条件。部分发布不等于 100% 产品覆盖。

当前置信度显示为 High（高）与 Low（低）两级。人工或 AI 审核身份保留在来源证据中；是否可以对客展示，由原有证据和冲突控制单独决定。详细定义见本次更新的数据口径手册。

info\_validity 仅表示政策已公布的截止日期；未知日期留空。freshness\_valid\_until 单独表示内部复核期限。用户已同意：全部 25 字段中，有记录的不适用与未公布状态计作完整；未知、无证据或待审核仍不完整。当前数字以带日期、源代码提交号和不可变执照校验值的最新发布审计为准。分别重新计算全部 25 字段记录状态完整度、日常 20 字段及全部 25 字段非空指标，不得将旧版本数字当作当前值；这些指标不证明政策准确率。

本次版本替换旧版双语 HTML/PDF 手册。旧版 PASS 标记、示例可用率、或未计时的恢复描述，不得用作当前验收证据。

## 安全边界

UFW 仅开放 22/80/443 端口；后端仅绑定本机回环地址，由 Caddy 对外提供服务。部署资料中的密钥以 backend/secrets.enc 加密保存（AES-256-CBC），并在部署主机解密；backend/.env 不进入 Git。手册不包含凭据。

## 信源修正与问题处置

发布前核对原始缓存、合并答案、线路、证据、按顺序加载的种子覆盖及运营覆盖六层资料，并重建准确的已审核候选。安装时暂停写入并重新核对；输入变化须重新审核。修正保留原始历史、问题及既有核验日期；变更字段记录实际且有范围限定的核验日期，不更新无关字段日期或期限，也不释放仍有独立问题的产品。

问题列表返回 revision\_sha256。客户端可提交 expected\_issue\_sha256，将屏幕审核绑定到具体版本。所有调用还受 API 自身读取至写入的版本检查；问题变化返回 HTTP 409，问题更新与 AuditEvent 同一事务提交。旧客户端省略预期版本时，不保护此前屏幕审核到 API 请求之间的变化。独立线路/证据预检不是线路记录的原子锁。POST 结果不确定时先查看当前状态，不盲目重试；保留实际操作者、说明和信源。

## 复核优先级

资格与续跑筛选后，先处理四条已报告的关键线路，再处理美国/香港至 18 个站点市场及中国内地的线路，然后其他站点至这些市场，之后处理其余库存。这是用户与项目优先级，不是实测流量。每第八次派发选择记录的信源复核尝试时间最早的合格线路；尝试日期缺失或无效者排在最前。这不以政策到期日排序，避免无限期搁置。续跑保留派发位置及原始五小时截止时间，发布不能重置预算或把未尝试内容计作核实。

## 验收证据

保留原始需求文件校验值和条款标识。发布报告区分已实现行为、实际测量和外部待办。计划任务不证明人工或供应商复核已发生；短时数据库恢复不证明完整服务恢复，短时监测不证明月度 99.99% 可用性。缺少的验收项目如实记录，不编造签署。

本文依据当前实现编制；最终验收仍需实际证据。